SOFIA MADSEN, PhD

Nationality - Swedish Residency - Switzerland 1991.10.04

CONTACT

sofia.b.madsen@gmail.com
+41 (0) 792815011
Lausanne, Switzerland
http://www.linkedin.com/in/smadsen1

LANGUAGES

Swedish	Mother Tongue
English	Fluent
French	Level B1/2

PUBLICATIONS

8 scientific publications

Link to all publications:
https://orcid.org/0000-0002-8980-2813

TEACHING

- Supervision of 3 students:

Premaster student, 3 months

Summer student, 3 months

Master student, 8 months

- Technical instructor for med. school practicals
- Training of new staff

SOFTWARES

Microsoft Office
Onedrive
Adobe
Affinity
Wrike
Sharepoint
R
ImageJ/Fiji
Prism

I have a robust scientific background with 10 years of experience in biomedical science. After almost 2 years of working with R&D and Preclinical Research in a start-up I am now eager to do a change in my career path. I have over the last year gained interest in the role of science in society, and how it can be made more available and accessible on a global level. Not only because the STEM fields are extremely interesting and fascinating, but also because they play important roles in understanding, developing and improving our surroundings and societies. This is why I am interested in the Science Corps Fellowship.

WORK EXPERIENCE

SCIENTIST, HAYA Therapeutics SA
2022-04 – 2023-12

PhD STUDENT, Knobloch lab, University of Lausanne
2017-06 – 2021-12

LAB. TECHNICIAN, DNA Section, Laboratory Medicine, Region Skåne
2017-02 – 2017-05

SCHOLAR, Wallenberg Neuroscience Center, Lund University
2016-01 – 2017-01

LAB. TECHNICIAN, Biobank, Laboratory Medicine, Region Skåne
Summer substitute 2014-05 – 2014-08 and 2015-07 – 2015-11

EDUCATION

Lemanic Neuroscience Doctoral School, University of Lausanne, Switzerland
2017 – 2021

Master in Biomedical Sciences, Lund University, Sweden
2013 – 2015

Exchange semester, Katholieke University Leuven, Belgium
2014

Bachelor in Biomedical Sciences, Lund University, Sweden
2010 – 2013

COURSES

Clinical Trials in Medicine, Gothenburg University, Sweden (remote)
2024-01 – 2024-05

SOFT SKILLS

Creative	Flexible	Goal Orientated
Positive	Teamwork	Time Management
Productive	Fast Learner	Project Management

SCIENTIFIC SKILLS

Neuroscience	Genetics	Preclinical Research
Stem Cell Biology	Pulmonary/Cardiac Fibrosis	*In Vitro* Experimentation
Metabolism	Molecular Biology	*In Vivo* Experimentation

SOFIA MADSEN, PhD

MAIN ACHIEVEMENTS

Scientist, HAYA Therapeutics SA

HAYA Therapeutics: Leveraging the power of the dark genome to develop precise and potent RNA therapies for reprogramming disease driving cell-states

- Preclinical research: disease model characterization, toxicology screenings, target validation and characterization.
- Establishing new methods for R&D and manage collaborations with contract research organizations (CROs).
- Data visualization and slide preparation for BoDs and Deep dives for external stakeholders.
- 7 week stay in San Diego, USA, to work with HAYAs Functional Genomics team.
- Highly engaged in setting up and implementing the electronic laboratory notebook Benchling.
- Improved Project Management and Communication skills from collaborative work across R&D teams and between HAYAs two sites (Lausanne and San Diego).
- Training of a BNF intern.

PhD student, Knobloch lab, Department of Biomedical Sciences, University of Lausanne

Laboratory of Stem Cell Metabolism: how metabolism regulates stem cell behaviour, primarily focused on neural stem cells

- Investigated the role of lipid droplets and lipid metabolism in the developing and adult mouse brain.
- Used the gene editing tool CRISPR/Cas9 to create and characterize a novel lipid droplet mouse model.
- Collaboration with CROs and Core Facilities.
- Supervised three students (a summer student, a pre-master student and a master student).
- Collaboration with Prof. Lutolf at EPFL on a project to characterize lipid droplets in intestinal organoids.
- Improved presentation and scientific communication skills through several poster presentations and talks at conferences. Moreover, awarded the price for the best poster at the Annual Retreat for the Department of Physiology 2019.

Scholar, Jakobsson Lab, Wallenberg Neuroscience Center, Lund University

Laboratory of Molecular neurogenetics: Transposable elements and how these viral-like sequences influence human brain evolution, neurodegenerative diseases, brain tumors and aging

- Used CRISP/Cas9 to to modify genes linked to neurodegenerative diseases.
- Characterization of a novel Adeno Associated Virus (AAV) based Huntington's Disease model.
- Investigating the mechanisms behind Huntington's Disease focusing on miRNA biogenesis and autophagy.

PRESENTATIONS

2021

- **Oral Presentation**- Lipid Droplet Workshop, Virtual
- **Poster Presentation** - International Society for Stem Cell Research (ISSCR), Virtual
- **Short Talk** - Annual meeting of the NeuroLeman Network and Doctoral Schools, Virtual

2019

- **Poster Presentation** - Eurogenesis, Bordeaux
- **Poster Presentation** - Lausanne Integrative Metabolism and Nutrition Alliance (LIMNA) Symposium, Lausanne
- **Poster Presentation** - Annual meeting of the NeuroLeman Network and Doctoral Schools, Diablerets
- **Best Poster Presentation Award** - Annual Retreat for Department of Physiology, Lausanne
- **Poster Presentation** - D-Day, Lausanne

2018

- **Poster Presentation** - Annual meeting of the NeuroLeman Network and Doctoral Schools, Diablerets